NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) Date:01-01-2025
Time: 3HRS GTM-8 Max. Marks: 300

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
B	D	A	B	B	D	C	B	D	D
11	12	13	14	15	16	17	18	19	20
D	A	D	D	C	A	B	A	D	B
21	22	23	24	25					
8	2	2	1	36					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
B	C	A	A	C	B	C	C	C	D
36	37	38	39	40	41	42	43	44	45
C	A	B	B	A	A	B	B	C	A
46	47	48	49	50					
50	62	3	14	6					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
D	C	A	D	A	C	A	A	D	D
61	62	63	64	65	66	67	68	69	70
A	A	D	B	A	B	B	A	B	A
71	72	73	74	75					
269	170	2	2	56					

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PHYSICS

1. x = 5 sin 10t

$$v_x = \frac{dx}{dt} = 50\cos 10t$$

Similarly $y = 5\cos 10t$

$$v_y = \frac{dy}{dt} = -50\sin 10t$$

$$v_{net}^2 = v_x^2 + v_y^2$$

$$v_{net} = \sqrt{(50)^2\left(\sin^2 10t + \cos^2 10t\right)} = 50 \text{ m/sec}$$

2. Density $\rho = \frac{m}{\pi r^2 L}$

$$\therefore \frac{\Delta\rho}{\rho}\times 100 = \left(\frac{\Delta m}{m} + 2\frac{\Delta r}{r} = \frac{\Delta L}{L}\right)\times 100$$

After substituting the values, we get the maximum percentage error in density = 4%.

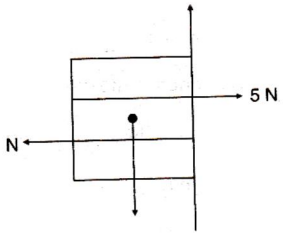
3. Component of velocities of both the particles in vertical direction are equal. Therefore, their time of flights are equal and their relative motion is in horizontal direction only. Thus, the maximum distance between them is equal to the difference in their horizontal ranges.

$$R_1 = \frac{(20)^2 \sin 60^o}{g} = \frac{(400)\sqrt{3}}{10\times 2} = 34.64\,m$$

$$R_2 = \frac{\left(\frac{20}{\sqrt{3}}\right)^2 \sin 120^o}{g} = \frac{400\times\sqrt{3}}{10\times 3\times 2} = 11.55\,m$$

$$\therefore S_{\max} = 36.64 - 11.55 = 23.09\,m = 23.1\,m$$

4. N = 6N



$(f)_{\max} = \mu\ N = (0.5)(5) = 2.5\,N$

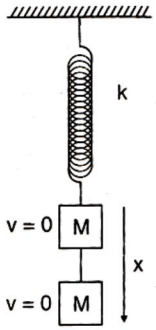
For vertical equilibrium of the block

$f = mg = 0.98 < (f)_{\max}$

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5.

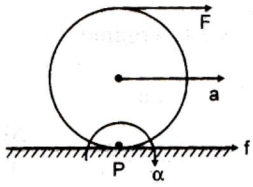


Let x be the maximum extension of the spring. From conservation of mechanical energy : decrease in gravitational potential energy
= increase in elastic potential energy

$\therefore \quad Mg\ x = \frac{1}{2}kx^2$

or $\quad x = \frac{2Mg}{k}$

6. Let f be the friction on the ring towards right a its linear acceleration and α the angular acceleration about centre of mass.



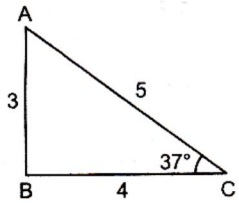
Point of contact P is momentarily at rest, i.e., ring will rotate about P.

$\therefore\ \alpha = \frac{\tau_P}{I_P} = \frac{F(2R)}{2MR^2} = \frac{F}{MR}$

$Now\ F + f = Ma = MR\alpha = F$

or $\quad f = 0$

7. AC > BC > AB



Moment of inertia of a body depends on distribution of mass from the axis of rotation. The mass is farthest from the axis AB. So, I_1 is maximum and nearest from the axis AC. So, I_3 is minimum or $I_3 < I_2 < I_1$.

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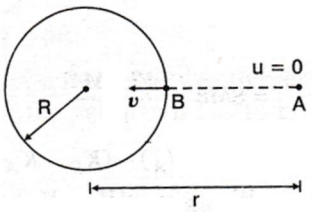
8. Conceptual

9. Let r be the radius of the satellite. Then

$$v_0^2 = \frac{GM}{r} \quad \ldots(1)$$

R = radius of earth

Applying conservation of mechanical energy between points A and B, we have :



$$\frac{1}{2}mv^2 = -\frac{GMm}{r} - \left(-\frac{GMm}{R}\right)$$

or $$v^2 = 2\frac{GM}{R} - \frac{2GM}{r}$$

$$= v_e^2 - 2v_o^2$$

or $$v = \sqrt{v_e^2 - 2v_o^2}$$

10. The displacement from the mean position to $\frac{A}{\sqrt{2}}$ and from $\frac{A}{\sqrt{2}}$ to A is same and $= \frac{T}{8}$.

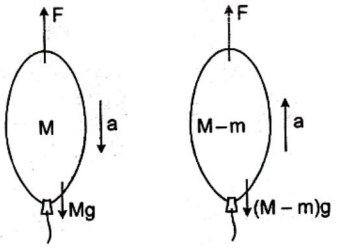
11. Upthrust F will remain same in both the cases.

Equations of motion will be :

$$Mg - F = Ma \quad \ldots(1)$$

and $$F - (M - m)g = (M - m)a \ldots(2)$$

Solving eqs. (1) and (2) we get, mass to be removed.



$$m = \frac{2Ma}{g + a}$$

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12. Length of the organ pipe is same in both the cases.

Fundamental frequency of open pipe is $f_1 = \frac{v}{2\ell}$ and frequency of third harmonic of closed pipe will be

$f_2 = 3\left(\frac{v}{4\ell}\right)$

Given that $f_2 = f_1 + 100$

or $f_2 - f_1 = 100$

or $\frac{3}{4}\left(\frac{v}{\ell}\right) - \left(\frac{1}{2}\right)\left(\frac{v}{\ell}\right) = 100$

$\Rightarrow \quad \frac{v}{4\ell} = 100\ Hz$

$\therefore \quad \frac{v}{2\ell}\ or\ f_1 = 200\ Hz$

Therefore, fundamental frequency of the open pipe is 200 Hz.

13. $\vec{E}$ is parallel to $\vec{B}$ *and* $\vec{v}$ is perpendicular to both. Therefore, path of the particle is a helix with increasing pitch. Speed of particle at any time t is

$v = \sqrt{v_x^2 + v_y^2 + v_z^2} \quad \ldots(1)$

Here, $v_y^2 + v_z^2 = v_0^2$

and $v_x^2 = \left(\frac{qE}{m}t\right)^2$

and $v = 2v_0$

Substituting the values in Eq. (1), we get

$t = \frac{\sqrt{3}mv_0}{qE}$

14. $4\ \Omega$ *and* $12\ \Omega$ resistances are in parallel. Therefore, their combined resistance is 3Ω. Similarly, 6Ω *and* 3Ω resistances are also in parallel and their combined resistance is 2Ω. Now the 3Ω *and* 2Ω resistances are in series. So the net resistance between A and B is $5\ \Omega$.

15. $I = I_{max}\cos^2\left(\frac{\phi}{2}\right)$

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$$\therefore \quad \frac{I_{max}}{4} = I_{max}\cos^2\frac{\phi}{2}$$

$$\cos\frac{\phi}{2} = \frac{1}{2}$$

or $$\frac{\phi}{2} = \frac{\pi}{3}$$

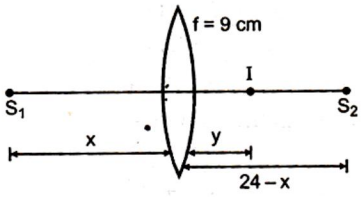
$$\therefore \quad \phi = \frac{2\pi}{3} = \left(\frac{2\pi}{\lambda}\right).\Delta x \qquad \ldots(1)$$

Where $\Delta x = d\sin\theta$

Substituting in Eq. (1), we get

$$\sin\theta = \frac{\lambda}{3d} \quad \text{or} \quad \theta = \sin^{-1}\left(\frac{\lambda}{3d}\right)$$

16. In this case one of the image will be real and the other virtual.
Let us assume that image of S_1 is real and that of S_2 is virtual.



Applying $$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

for S_1 :

$$\frac{1}{y} + \frac{1}{x} = \frac{1}{9} \qquad \ldots(1)$$

for S_2 :

$$-\frac{1}{y} + \frac{1}{24-x} = \frac{1}{9} \qquad \ldots(2)$$

Solving Eqs.(1) and (2), we get
x = 6 cm

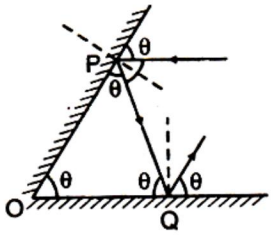
17. $\angle OPQ = \angle OQP = \theta$

or $3\theta = 180^o$

$\therefore \quad \theta = 60^o$

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18. 1 calorie is the heat required to raise the temperature of 1 g of water from 14.5°C to 15.5°C at 760 mm of Hg.

19. In SHM $v_{max} = a\omega$

Here, a = amplitude and ω = angular velocity

$$\therefore \quad < v >= \frac{\text{distance travelled in one oscillation}}{\textit{time} \text{ period}}$$

$$= \frac{4a}{T} = \frac{4a}{\frac{2\pi}{\omega}}$$

$$= \frac{2a\omega}{\pi} = \frac{2v_{max}}{\pi}$$

20. $$T = \frac{d}{\frac{v}{\sqrt{2}}} + \frac{d}{\frac{ev}{\sqrt{2}}} = \left(1 + \frac{1}{e}\right)\frac{\sqrt{2}d}{v}$$

or $$\frac{\frac{2v}{\sqrt{2}}}{g} = \left(1 + \frac{1}{e}\right)\frac{\sqrt{2}d}{v}$$

or $$e = \frac{gd}{v^2 - gd}$$

21. Equating the rate of flow of liquid

$$Q = \frac{\pi r^4 P}{8\eta\ell}$$

$$\therefore \quad Q_1 = Q_2$$

$$\therefore \quad \frac{\pi(r)^4 P_1}{8\eta\ell} = \frac{\pi(2r)^4 P_2}{8\eta(2\ell)}$$

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$\therefore \quad \frac{P_1}{P_2} = 8$

22. $T \propto \frac{n^3}{Z^2}$

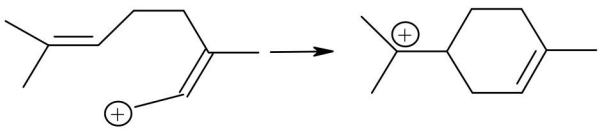
23. $V_{RMS} = \frac{V_0}{\sqrt{2}}$ for full wave rectifier and $\frac{V_0}{2}$ for half wave rectifier.

24. The freezing time is independent of surface area of lake.

25. The common temperature = $\frac{\sum msT}{\sum ms}$

CHEMISTRY

26.



27. Compound with higher R_f value having high solubility in mobile phase hence less adsorbed.

28. $r = K[conc]^n$ (n = order)
Slope of graph (n) = D > B > A > C
log (rate) = log k + n log [conc]

29. $Tb = [Xe]4f^9 6s^2$
$Eu : [Xe]4f^7 6s^2$

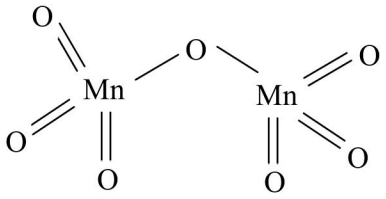
30. CCl_4 non polar but contains polarbonds

31. $pH = pKa + \log\left(\frac{salt}{acid}\right)$
$= 4.76 + \log\left(\frac{1/100}{1/100}\right) = 4.76$

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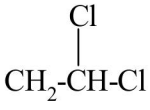
32.



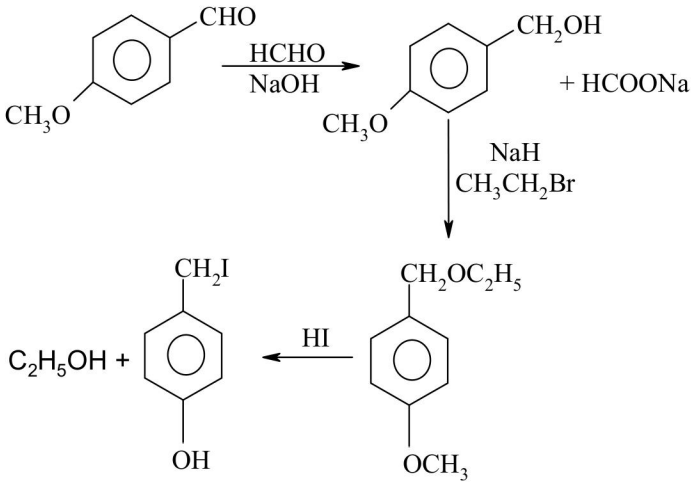
33. $CN^{\ominus}$ is comparatively strong ligard.

34. Intramolecular aldol condensation

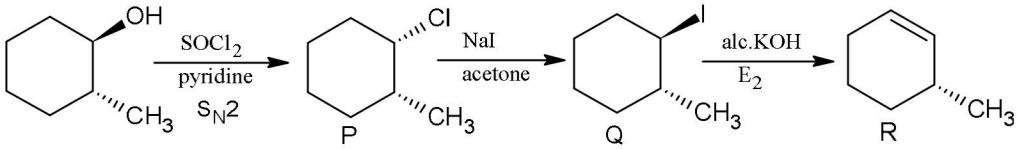
35.



36.



37.



38. Conceptual

39. $X_{(s)} \rightleftharpoons A_{(g)} + 2B_{(g)}$

x 2x+2y

$Y_{(s)} \rightleftharpoons C_{(g)} + 2B_{(g)}$

y 2x+2y

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$$\frac{K_{P_1}}{K_{P_2}} = \frac{x}{y} \Rightarrow x = 2y$$

$$K_{P_1} = x(2x + 2y)^2 \Rightarrow x = 0.1 \text{ atm}, \ y = 0.05 \text{ atm}$$

$$P_T = 3(x + y) = 0.45 \text{ atm}$$

40. 90 g Hg has 10 g Na

$$\therefore \ 10 \text{ g Hg} = \frac{10}{9} \text{g Na}$$

$$w_{Na} = \frac{m}{n} \times \frac{i \times t}{96500}$$

$$\frac{10}{9} = \frac{23}{1} \times \frac{10 \times t}{96500}$$

$t = 466.18$ sec $= 7.77$ min

41. $260 = \Lambda_m^\infty - 0.5B$ … (1)

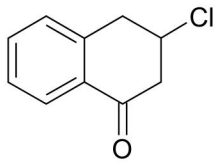
$250 = \Lambda_m^\infty - B$ …(2)

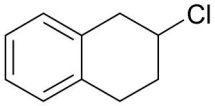
On solving (1) and (2), we get

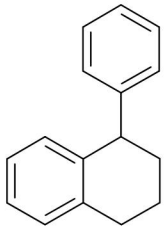
$\Lambda_m^\infty = 270 \ \Omega^{-1}cm^2mol^{-1}$

42. conceptual

43.

A = 

B = 

C = 

44. For acidic amino acid ; $P^I = \frac{1.88 + 3.65}{2}$

For basic amino acid; $P^I = \frac{8.95 + 10.53}{2}$

45. conceptual

46. M.eq. of $K_2Cr_2O_7$ = M.eq.of FeC_2O_4

$$FeC_2O_4 + Cr_2O_7^{2-} \rightarrow Fe^{3+} + CO_2 + Cr^{3+}$$

$$V \times 0.02 \times 6 = \frac{0.288}{144} \times 3 \times 1000; \ V = 50 \text{ mL}$$

47. Given n = 3

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$T_1 = 300;\ T_2 = 1000$

$C_p = 23 + 0.01T$

The relation between ΔH and C_p is

$$\Delta H = \int_{T_1}^{T_2} nC_p dT \qquad \ldots(i)$$

After putting all variable values is eq. (i)

$$\Delta H = \int_{300}^{1000} (23 + 0.01T)dT \qquad = 3\left[23T + \frac{0.01T^2}{2}\right]_{300}^{1000}$$

$$= 3\left[16100 + 4550\right] = 3 \times 20650 = 61950\,J \ = 61.95\ kJ$$

48. Only Fullerenes, White phosphorus and Rhombic sulphur contain discrete molecules.

49. $4f^{14}, 5f^0$

50. $\left[Pt(NH_3)_2(SCN)_2\right]$ cis, trans

$\left[Pt(NH_3)_2(NCS)(SCN)\right]$ cis, trans

$\left[Pt(NH_3)_2(NCS)_2\right]$ cis, trans

Complex is square planar.

MATHEMATICS

51. Given circle $x^2 + y^2 - 2gx + 6y - 19c = 0$

Passes through (6, 1)

$12g + 19c = 43 \qquad \ldots(i)$

Centre (g, -3) lies on given line

So, $g + 6c = 8 \qquad \ldots(ii)$

Solve equation (i) & (ii)

$\Rightarrow$ $c = 1$ & $g = 2$ equation of circle $x^2 + y^2 - 4x + 6y - 19 = 0$

x-intercept $= 2\sqrt{g^2 - c} = 2\sqrt{23}$

52. Given expression is

$$= \lim_{n\to\infty} \frac{0 + 3 + 6 + 9 + ...n\ terms}{\sqrt{2n^4 + 4n + 3} - \sqrt{n^4 + 5n + 4}}$$

$$\Rightarrow \lim_{n\to\infty} \frac{3n(n-1)}{2\left(\sqrt{2n^4 + 4n + 3} - \sqrt{n^4 + 5n + 4}\right)}$$

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$$\Rightarrow \lim_{n\to\infty} \frac{3\left(1-\frac{1}{n}\right)}{2\left(\sqrt{2+\frac{4}{n^3}+\frac{3}{n^4}}-\sqrt{1+\frac{3}{n^3}+\frac{4}{n^4}}\right)}$$

Apply the limit

$$=\frac{3}{2\left(\sqrt{2}-1\right)}=\frac{3}{2}\left(\sqrt{2}+1\right)$$

53. $\tan^{-1}\left(\frac{1}{1+n(n+1)}\right)=\tan^{-1}(n+1)-\tan^{-1}(n)$

$$\Rightarrow \sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)=\left(\tan^{-1}2-\tan^{-1}1\right)+\left(\tan^{-1}3-\tan^{-1}2\right)+\ldots+\left(\tan^{-1}51-\tan^{-1}50\right)$$

$$=\tan^{-1}51-\tan^{-1}1$$

$$=\tan^{-1}\left(\frac{50}{52}\right)=\cot^{-1}\left(\frac{52}{50}\right)$$

54. Given $A^2=3A+\alpha I \Rightarrow A^3=3A^2+\alpha A$

$\Rightarrow A^3=3(3A+\alpha I)+\alpha A$ [since $A^2=3A+\alpha I$]

$\Rightarrow A^3=9A+\alpha A+3\alpha I$

$A^4=(9+\alpha)A^2+3\alpha A=(9+\alpha)(3A+\alpha I)+3\alpha A$

$\Rightarrow 21A+\beta I=A(27+6\alpha)+\alpha(9+\alpha)\mathrm{I}$

$\Rightarrow 27+6\alpha=21\Rightarrow\alpha=-1\Rightarrow\beta=\alpha(9+\alpha)=-8$

55. Given system has infinitely many solution.

$$\Rightarrow\begin{vmatrix}2 & 1 & -1\\ 2 & -5 & \lambda\\ 1 & 2 & -5\end{vmatrix}=0$$

$\Rightarrow 2(25-2\lambda)-(-10-\lambda)-(4+5)=0$

$\Rightarrow 51=3\lambda\Rightarrow\lambda=17$

and $\begin{vmatrix}2 & 1 & 5\\ 2 & -5 & \mu\\ 1 & 2 & 7\end{vmatrix}=0$

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$\Rightarrow 2(-35-2\mu)-(14-\mu)+5(4+5)=0$

$-3\mu=39 \Rightarrow -\mu=13$

$(\lambda+\mu)^2+(\lambda-\mu)^2=2(\lambda^2+\mu^2)=2(17^2+13^2)=916$

56. $f(x)=\lim\limits_{n\to\infty}\left((2x)^n+(x^2)^n\right)^{1/n}$

$=\max(2^x,x^2)$ and $f(x)$ is even function.

57. $c^2>a^2+b^2 \Rightarrow \angle C>90° \Rightarrow \angle A,\angle B<45°$

$f(x)=\sin^5 x+\cos^5 x$

$f'(x)=5.\sin x.\cos x(\sin^3 x-\cos^3 x) \Rightarrow f'(x)<0 \forall x \in \left(0,\frac{\pi}{4}\right)$

58. Given integral is

$I(x)=\int \sec^2 x \sin^{-2022} x dx-2022\int \sin^{-2022} x dx$

$=\tan x(\sin x)^{-2022}+\int (2022)\tan x(\sin x)^{-2023}\cos x\, dx$

$-2022\int (\sin x)^{-2022} dx$

$I(x)=(\tan x)(\sin x)^{-2022}+C \quad \ldots(i)$

$At\ x=\frac{\pi}{4}, 2^{1011}=\left(\frac{1}{\sqrt{2}}\right)^{-2022}+C \Rightarrow C=0$

Hence $I(x)=\frac{\tan x}{(\sin x)^{2022}}$

Put $x=\frac{\pi}{6}$ in the above expression,

$I\left(\frac{\pi}{6}\right)=\frac{1}{\sqrt{3}\left(\frac{1}{2}\right)^{2022}}=\frac{2^{2022}}{\sqrt{3}}$

$I\left(\frac{\pi}{3}\right)=\frac{\sqrt{3}}{\left(\frac{\sqrt{3}}{2}\right)^{2022}}=\frac{2^{2022}}{\left(\sqrt{3}\right)^{2021}}=\frac{1}{3^{1010}}I\left(\frac{\pi}{6}\right)$

$3^{1010}I\left(\frac{\pi}{3}\right)=I\left(\frac{\pi}{6}\right)$

59. Given, $f(x)+f(\pi-x)=\pi^2$

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Let $I=\int_0^{\pi} f(x)\sin x dx$

$I=\int_0^{\pi} f(\pi-x)\sin(\pi-x)dx$ [applying King's Rule]

$\Rightarrow 2I=\int_0^{\pi}[f(x)+f(\pi-x)]\sin x dx$

$2I=\int_0^{\pi}\pi^2\sin x dx \Rightarrow 2I=\pi^2\int_0^{\pi}\sin x dx$

$2I=\pi^2\times 2 \Rightarrow I=\pi^2$

60. $\rightarrow abc+ab+a=a(1+b(c+1))$

$\rightarrow P(3k)=P(3k+1)=P(3k+2)=1/3$

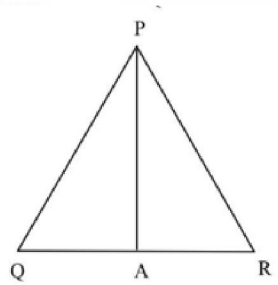
(i) a is divisible by $3 \Rightarrow \frac{1}{3}=\frac{9}{27}$

(ii) a is not divisible by 3 and $b(c+1)$ is $3k+2$ type

$\Rightarrow \frac{2}{3}\left[\frac{1}{3}\times\frac{1}{3}\times 2\right]=\frac{4}{27}$

$P=\frac{13}{27}$

61.



$PA=\frac{|\overline{PQ}\times\overline{PR}|}{|\overline{QR}|}=\frac{3\sqrt{26}}{7}$

62. $h(x)=\frac{f(x)-f(-x)}{2}=\frac{x}{2}$

63. Let α *and* β be the roots of the quadratic equation

$7x^2-3x-2=0$

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$\therefore \alpha+\beta=\frac{3}{7}, \alpha\beta=\frac{-2}{7}$

Now, $\frac{\alpha}{I-\alpha^2}+\frac{\beta}{I-\beta^2}$

$=\frac{\alpha-\alpha\beta(\alpha+\beta)+\beta}{1-(\alpha^2+\beta^2)+(\alpha\beta)^2}$

$=\frac{(\alpha+\beta)-\alpha\beta(\alpha+\beta)}{1-(\alpha+\beta)^2+2\alpha\beta+(\alpha\beta)^2}$

$=\frac{\frac{3}{7}+\frac{2}{7}\times\frac{3}{7}}{1-\frac{9}{49}+2\times\frac{-2}{7}+\frac{4}{49}}=\frac{27}{16}$

64. Given $\left\{\frac{4^{2022}}{15}\right\}=\left\{\frac{16^{1011}}{15}\right\}$

$=\left\{\frac{(1+15)^{1011}}{15}\right\}=\left\{\frac{15k+1}{15}\right\}=\frac{1}{15}$

65. Given , $T_4 = 500$. Let first term = a

r = common ratio $=\frac{1}{m}, m\in N$,

Then, $ar^3=500$

$\Rightarrow\frac{a}{m^3}=500$

$S_n-S_{n-1}=ar^{n-1}\Rightarrow S_6>S_5+1\Rightarrow S_6-S_5>1$

$ar^5>1\Rightarrow\frac{500}{m^2}>1$

$m^2<500$ …(i)

and $S_7-S_6<\frac{1}{2}$

$\Rightarrow\frac{a}{m^6}<\frac{1}{2}$

$m^3>10^3\Rightarrow m>10$ …(ii)

From (i) and (ii)

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m = 11, 12, 13, ……22

66. Since, slope of reflected ray $= \tan 60^o = \sqrt{3}$ and equation ray $y = \frac{x}{\sqrt{3}}$ intersect y + x = 1 at

$\left(\frac{\sqrt{3}}{\sqrt{3}+1}, \frac{1}{\sqrt{3}+1}\right)$

Equation of reflected ray is $y - \frac{1}{\sqrt{3}+1} = \sqrt{3}\left(x - \frac{\sqrt{3}}{\sqrt{3}+1}\right)$

For abscissa

Put $y = 0 \Rightarrow x = \frac{2}{3+\sqrt{3}}$

67. Equation of ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \ a > b$.

Given eccentricity is $\frac{1}{4}$.

$\Rightarrow e^2 = 1 - \frac{b^2}{a^2}$

$\frac{1}{16} = 1 - \frac{b^2}{a^2} \Rightarrow \frac{b^2}{a^2} = 1 - \frac{1}{16} = \frac{15}{16} \Rightarrow b^2 = \frac{15}{16}a^2$

Put the value of b^2 in equation of ellipse and satisfy the point $\left(-4\sqrt{\frac{2}{5}}, 3\right)$

$\Rightarrow \frac{16 \times \frac{2}{5}}{a^2} + \frac{9}{b^2} = 1$

$\Rightarrow \frac{32}{5a^2} + \frac{9}{b^2} = 1$

$\Rightarrow \frac{32}{5a^2} + \frac{9}{\frac{15}{16}a^2} = 1 \Rightarrow \frac{80}{5a^2} = 1 \Rightarrow 16 = a^2$

Put the value of a^2 in $b^2 \Rightarrow b^2 = \frac{15}{16}a^2$

$b^2 = 15$

Add a^2 & b^2, $a^2 + b^2 = 16 + 15 = 31$

68. Since, (a, b) R (c, d)

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$\Rightarrow ad(b-c)=bc(a-d)$

For symmetric :

(c, d) R (a, b) $\Rightarrow$ cb (d – a) = da(c – b)

$\Rightarrow$ (a, b) R (c, d) symmetric

For reflexive :

(a, b) R(a, b) $\Rightarrow$ ab(b – a) $\neq$ ba(a – b) $\Rightarrow$Not reflexive and for Transitive :

(2, 3) R(3, 2) and (3, 2) R(5, 30) but $\big((2,3),(5,30)\big)\notin R \Rightarrow$ Not transitive.

69. $g(x)=x \text{ if } x \text{ is even}$

if x is odd g(x) can take any value in A

$\Rightarrow 1\times 10^5$

70. $|A|=2[3]-1[2]=4$

$\therefore \quad \left|adj\left(adj\left(adj2A\right)\right)\right|$

$\therefore |2A|^{(n-1)^3} \Rightarrow |2A|^8=16^n \qquad [\because$ n = order of matrix = 3]

$\Rightarrow \left(2^3|A|\right)^8=16^n \qquad \left[\because |kA|=(k)^n|A|\right]$

$\Rightarrow \left(2^3\times 2^2\right)^8=16^n \Rightarrow 2^{40}=16^n \Rightarrow 16^{10}=16^n$

$\Rightarrow n=10$

71. Given, $\overline{x}=50$

$\Rightarrow \sum x_i=500$

Now, $\sum x_{i_{correct}}=500+20+25-45-50=450$

Since, $\sigma^2=144$

$\Rightarrow \dfrac{\sum x_i^2}{10}-(50)^2=144$

$\sum x_{i_{correct}}^2=\left(144+(50)^2\right)\times 10-(45)^2-(50)^2+(20)^2+(25)^2=22940$

Correct variance $=\dfrac{\sum\left(x_{i_{correct}}\right)^2}{10}-\left(\dfrac{\sum x_{i_{correct}}}{10}\right)^2$

$= 2294 – (45)^2 = 2294 – 2025 = 269$

72. Given curve is $4x^3 – 3xy^2 + 6x^2 – 5xy – 8y^2 + 9x + 14 = 0$

Differentiate both sides

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$12x^2-3y^2-6xyy'+12x-5y-5xy'-16yy'+9=0$

Satisfy the point (-2, 3),

$\Rightarrow 48-27+36y'-24-15+10y'-48y'+9=0$

$\Rightarrow 2y'=-9$

So, $m_1=\frac{-9}{2}$ & $m_2=\frac{2}{9}$

Equation of tangent & normal is shown below

$T\equiv y-3=\frac{-9}{2}(x+2)$

Put, y = 0

$x=\frac{-4}{3}$

$N\equiv y-3=\frac{2}{9}(x+2)$

Put, y = 0 $\Rightarrow x=\frac{-31}{2}$

Therefore, area $=\frac{1}{2}\times Base\times Height$

$A=\frac{1}{2}\times\left(\frac{-4}{3}+\frac{31}{2}\right)(3)=\frac{1}{2}\left(\frac{85}{6}\right)\cdot 3=\frac{85}{4}$

8A = 170

73. Given that

$(x\cos x)dy+(xy\sin x+y\cos x-1)dx=0,\ 0<x<\frac{\pi}{2}$

$\frac{dy}{dx}+\left(\frac{x\sin x+\cos x}{x\cos x}\right)y=\frac{1}{x\cos x}$

I.F. = x sec x

Then solution is $y.x\sec x=\int\frac{x\sec x}{x\cos x}dx=\tan x+c$

Since $y\left(\frac{\pi}{3}\right)=\frac{3\sqrt{3}}{\pi}$, $we\ get\ c=\sqrt{3}$

Hence $\left|\frac{\pi}{6}y''\left(\frac{\pi}{6}\right)+y'\left(\frac{\pi}{6}\right)\right|=|-2|=2$

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74. $z^2+z+1=0 \Rightarrow z=w,\ w^2$

$$\left|\sum_{n=1}^{15}\left(z^n+(-1)^n\frac{1}{z^n}\right)^2\right|$$

On squaring we get

$$=\left|\sum_{n=1}^{15}\left(z^{2n}+\frac{1}{z^{2n}}+2(-1)^n\right)\right|=\left|\sum_{n=1}^{15}w^{2n}+\frac{1}{w^{2n}}+2(-1)^n\right|$$

Using G.P. we get

$$=\left|\frac{w^2\left(1-w^{30}\right)}{1-w^2}+\frac{\frac{1}{w^2}\left(1-\frac{1}{w^{30}}\right)}{1-\frac{1}{w^2}}+2(-1)\right|$$

$$=\left|\frac{w^2(1-1)}{1-w^2}+\frac{\frac{1}{w^2}(1-1)}{1-\frac{1}{w^2}}-2\right|$$

$$=|0+0-2|=2$$

75. Total 16 cubes are given and out of which 11 are blue and 5 are red.

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

$x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6$

$x_1+x_2+x_3+x_4+x_5+x_6=11$

$x_1, x_6 \geq 0,\ x_2, x_3, x_4, x_5 \geq 2$

$x_2=t_1+2 \Rightarrow x_3=t_3+2 \Rightarrow x_4=t_4+2 \Rightarrow x_5=t_5+2$

$x_1, t_2, t_3, t_4, t_5, x_6 > 0$

No. of solutions $={}^{6+3-1}C_3={}^{8}C_3=56$

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